\# Nifty100 Financial Intelligence Platform

\## Analyst User Guide

\### 1. Overview

The Nifty100 Financial Intelligence Platform is a financial analysis system covering 92 Nifty 100 companies.

It combines financial statements, calculated financial ratios, cash-flow analytics, valuation metrics, peer comparisons, sector analysis, clustering, and automated insights into a single platform.

The project uses SQLite for structured storage and Python-based analytics modules for processing and analysis.

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\## 2. Data and Analytics Flow

The general workflow is:

Raw Financial Data

  ↓

ETL and Data Validation

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SQLite Database

  ↓

Financial KPI Calculation

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Screening / Peer / Sector / Valuation Analysis

  ↓

Reports and REST API

The database contains company information, financial statements, ratios, market data, peer groups, sectors, documents, analysis results, and generated insights.

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\## 3. Financial Analysis

The platform calculates and exposes multiple financial indicators, including:

\- Net Profit Margin

\- Operating Profit Margin

\- Return on Equity (ROE)

\- Debt to Equity

\- Interest Coverage

\- Asset Turnover

\- Free Cash Flow

\- Capital Expenditure

\- Earnings per Share

\- Dividend Payout Ratio

\- Revenue CAGR

\- PAT CAGR

\- EPS CAGR

\- Cash-flow quality indicators

These metrics can be used to evaluate profitability, leverage, operating efficiency, growth, and cash generation.

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\## 4. Financial Screener

The screener allows companies to be filtered using financial thresholds.

Examples include:

\- Minimum ROE

\- Maximum Debt/Equity

\- Minimum Free Cash Flow

\- Revenue CAGR

\- PAT CAGR

\- Operating Margin

\- P/E

\- P/B

\- Dividend Yield

\- Interest Coverage

Preset screening strategies are also available for common investment-analysis use cases.

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\## 5. Company Analysis

Individual companies can be analysed through their:

\- Profit and Loss history

\- Balance Sheet history

\- Cash Flow history

\- Financial ratios

\- Market-cap data

\- Documents

\- Peer comparison

\- Company tear sheet

The platform supports historical financial analysis rather than relying only on a single period.

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\## 6. Peer and Sector Analysis

Companies can be compared with their relevant peer groups.

Peer analysis supports comparison of financial metrics and helps identify companies performing above or below their peer group.

Sector analysis provides aggregated financial information across sectors and can be used to understand differences in profitability, growth, leverage, and valuation.

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\## 7. Clustering

The analytics layer also groups companies using financial characteristics.

The clustering model uses:

\- Return on Equity

\- Debt to Equity

\- Operating Profit Margin

\- Revenue CAGR

\- 5-year Free Cash Flow CAGR

K-Means clustering is used to identify groups of companies with similar financial characteristics.

Supporting outputs include:

\- Cluster labels

\- Cluster profiles

\- Cluster outliers

\- Elbow plot

\- Feature correlation heatmap

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\## 8. Valuation Analysis

The valuation module provides indicators such as:

\- Free Cash Flow Yield

\- Sector Median P/E

\- P/E versus Sector Median

\- Valuation Flags

Companies can be classified using valuation signals such as:

\- Fair

\- Discount

\- Caution

The analysis is intended to support comparison rather than replace investment judgement.

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\## 9. Cash Flow Intelligence

Cash-flow analysis evaluates:

\- Free Cash Flow

\- CFO quality

\- CapEx intensity

\- FCF conversion

\- Capital allocation behaviour

\- Distress signals

\- Debt-funded growth

\- Cash accumulation patterns

These indicators provide additional context beyond accounting profitability.

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\## 10. NLP Insights

The platform includes an NLP component for generating company-level pros and cons from available financial analysis.

Generated insights can be used as a quick summary before performing deeper fundamental analysis.

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\## 11. REST API

The platform exposes financial intelligence through a REST API.

Current API areas include:

\- Health check

\- Company listing

\- Company details

\- Profit and Loss

\- Balance Sheet

\- Cash Flow

\- Financial ratios

\- Company tear sheet

\- Screener

\- Sector analysis

\- Sector companies

\- Peer groups

\- Peer comparison

\- Market-cap data

\- Portfolio statistics

\- Company documents

The API provides programmatic access to the same underlying financial intelligence.

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\## 12. Portfolio Statistics

The portfolio statistics endpoint provides distribution statistics for major financial indicators.

For each metric it can provide:

\- P10

\- P25

\- P50

\- P75

\- P90

\- Mean

\- Standard Deviation

This allows analysts to understand the distribution of financial characteristics across the covered companies.

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\## 13. Reports and Outputs

Important generated outputs include:

\- Screener results

\- Peer comparison reports

\- Valuation summaries

\- Valuation flags

\- Capital allocation analysis

\- Cash-flow intelligence

\- Cluster labels

\- Cluster profiles

\- Cluster outliers

\- Portfolio statistics

\- Company tear sheets

\- Sector reports

\- Radar charts

\- API test reports

\- Performance validation notes

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\## 14. API Validation

The REST API has been validated using automated integration tests.

The API test suite covers important endpoints including:

\- Health

\- Companies

\- Company details

\- Ratios

\- Screener

\- Sectors

\- Portfolio statistics

\- Market cap

\- Documents

The complete regression suite currently passes successfully.

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\## 15. Performance Validation

API performance smoke tests cover repeated requests against:

\- Health endpoint

\- Company endpoint

\- Screener endpoint

The checks verify that the endpoints return successful responses within the configured validation threshold.

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\## 16. Testing

The project uses pytest for automated testing.

Testing covers:

\- ETL

\- Data normalisation

\- CAGR calculations

\- Cash-flow KPIs

\- Financial ratios

\- API integration

\- API performance

All implemented test groups should be run before final submission.

Command:

  pytest -q

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\## 17. Running the Project

\### Create virtual environment

  python -m venv venv

\### Activate on Windows

  venv\\Scripts\\activate

\### Install dependencies

  pip install -r requirements.txt

\### Run tests

  pytest -q

\### Run the REST API

  python -m uvicorn src.api.main:app --port 8001

\### API documentation

Once the API is running, the interactive API documentation is available through the FastAPI documentation endpoint.

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\## 18. Analyst Workflow

A typical analyst workflow is:

1\. Check the API/database health.

2\. Review the company or sector.

3\. Examine profitability and leverage ratios.

4\. Review historical growth and cash flow.

5\. Compare the company with peers.

6\. Check valuation indicators.

7\. Review generated pros and cons.

8\. Use clustering and portfolio statistics for additional context.

9\. Export or review the relevant reports.

10\. Perform final analyst judgement using multiple indicators.

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\## 19. Important Note

The platform is designed as a financial intelligence and analysis tool.

Screening, valuation flags, clustering, and generated insights should be treated as analytical aids. Final investment decisions should not be based on a single metric or automated signal.